



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section:M

Group No: 00

Project Report On

Project Name: NeedBridge

Supervised By

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Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of	Demonstrates limited understanding of verification processes and data flow in the system. Verification	Shows a basic understanding of verification processes and attempts to verify system data. However, verification	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification

	understanding or implementation of verification processes, and data flow is not considered.	attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	thorough, with attention to ensuring data integrity and appropriate data flow within the system.	efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a

		or can't be identified as coherent.	described as user friendly.	friendly coherent system.	preferred way according to the coherent usage .
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NeedBridge

Introduction

NeedBridge is a multi-vendor e-commerce desktop application built using C# Windows Forms and Microsoft SQL Server. The platform bridges the gap between customers and vendors by providing a centralized marketplace where vendors can list products and customers can browse, purchase, and review them similar to how Daraz or Bikroy operates in Bangladesh. The system is managed a SuperAdmin at the top level, supported by Admins who handle daily operations, while Vendors manage their own storefronts and Customers enjoy a full shopping experience all within a single integrated platform.

User Story

Super Admin

The SuperAdmin is the highest authority on the NeedBridge platform with complete control over everything. After logging in with their credentials, the SuperAdmin can create new admin accounts to assign platform managers, and suspend, activate, or permanently delete those accounts when needed. They can monitor the entire platform through a dashboard showing total admins, vendors, customers, orders, and revenue at a glance. They also have direct access to manage vendors and customers in case of escalated issues, and can view the full platform sales report to oversee revenue performance. All actions are protected by a secure logout system.

Admin

The Admin is responsible for the day-to-day management and smooth operation of the NeedBridge platform. After logging in, they access a dashboard showing platform-wide statistics and recent orders. Admins can view and manage vendor accounts — suspending or activating them based on policy compliance — and do the same for customer accounts. They have the authority to review and delete inappropriate or fake reviews to maintain platform quality. When disputes arise between customers and vendors, Admins can update or cancel any order to resolve the conflict. They can also set or remove discount percentages on any product to run promotions, monitor stock levels across all vendors, and generate sales reports filtered by time period to track overall platform performance.

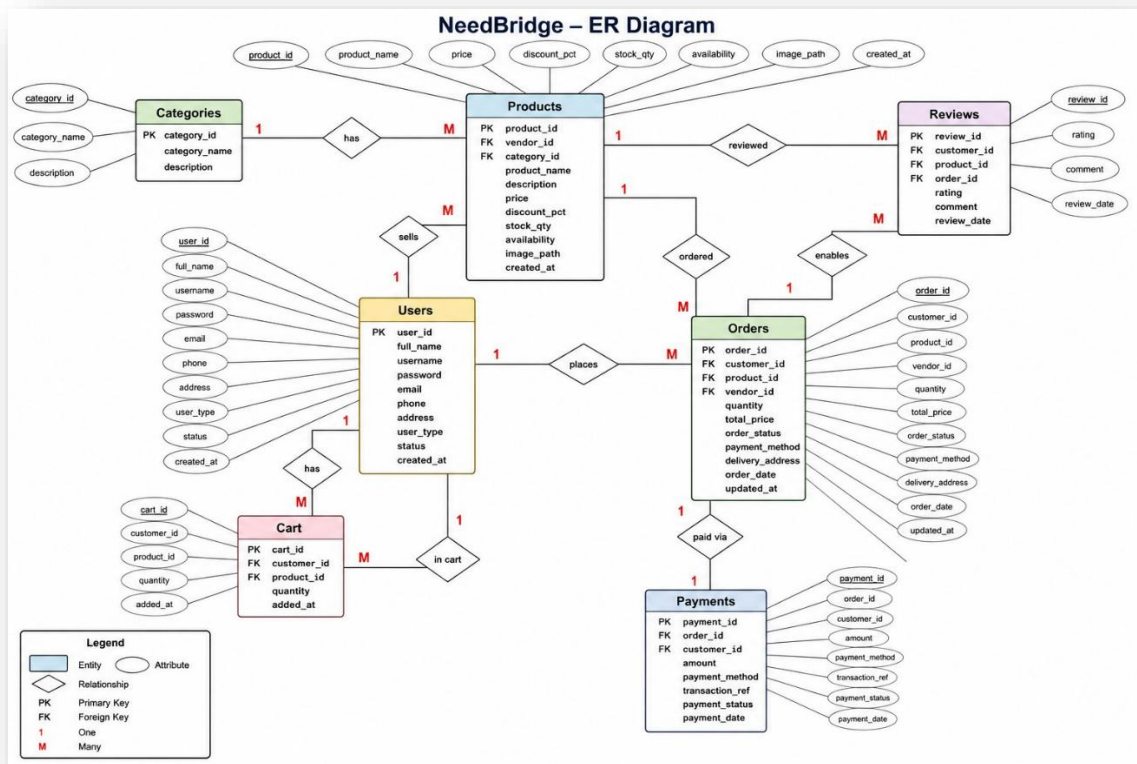
Vendor

The Vendor is an independent seller on the NeedBridge platform who manages their own storefront. After registering and logging in, they are greeted by a dashboard summarizing their total products, orders, earnings, and any low stock alerts. Vendors can add new products with full details including name, description, category, price, discount percentage, stock quantity, and an optional image. They can edit or delete their listings at any time to keep their store up to date. When customers place orders, vendors can view them and progressively update the delivery status from Confirmed through to Delivered. A dedicated stock dashboard helps them identify which products are running low, while the earnings report — filterable by day, week, month, or all time — gives a clear picture of their revenue and best-selling products.

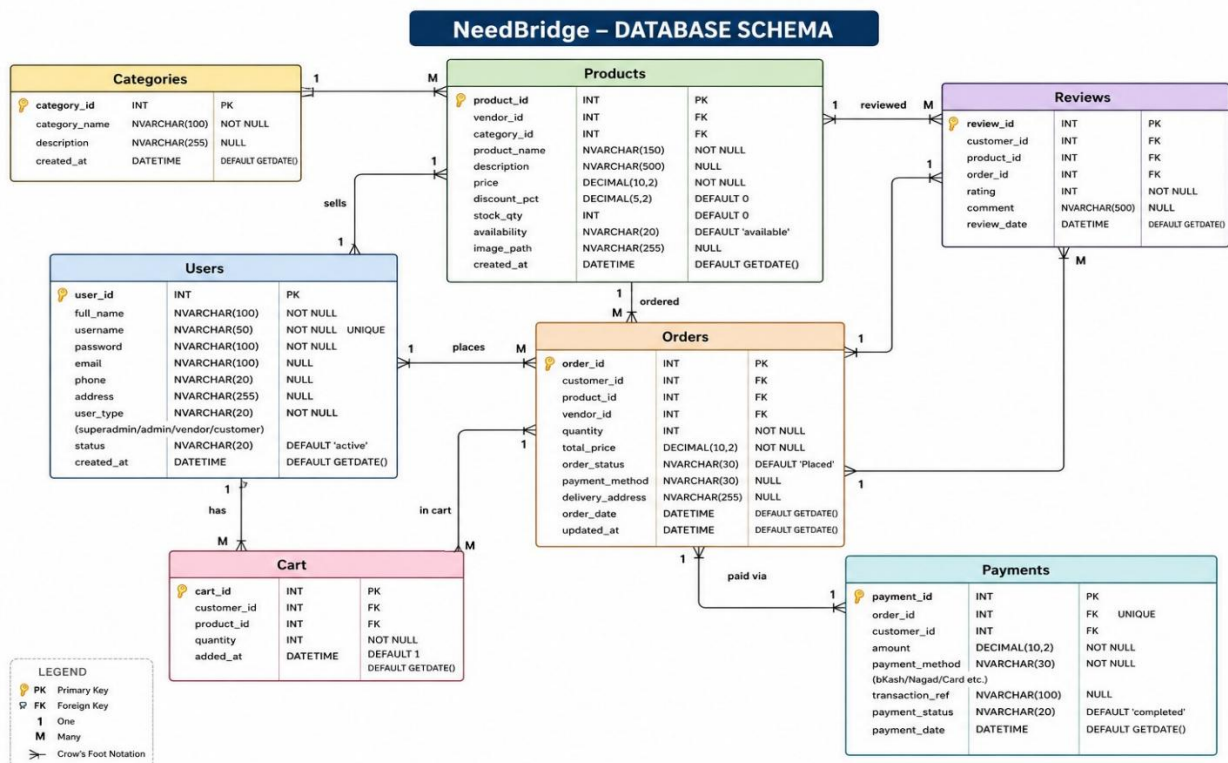
Customer

The Customer is the end user of the NeedBridge platform who browses, selects, and purchases products from various vendors. After registering and logging in, they land on the home page where all available products are displayed with the ability to search by name and filter by category and price range. Clicking on any product opens the full details page showing the description, price, any active discount, stock availability, and reviews left by previous buyers. Customers can add items to their cart for a combined checkout or use the Buy Now option for a faster single-item purchase. Payment is supported via bKash, Nagad, Rocket, Card, or Cash on Delivery in BDT. After placing an order, customers can track its progress through the My Orders section, and cancel it if it has not yet shipped or request a return if already delivered. Once a purchase is complete, they can leave a star rating and written review for the product. Their profile page allows them to update personal information and change their password at any time.

Er-Diagram:-



Database Schema:-



SQL Queries:-

DATABASE-

```
CREATE DATABASE NeedBridgeDB;  
GO
```

```
USE NeedBridgeDB;  
GO
```

Users Table –

```
CREATE TABLE Users (  
  
    user_id INT PRIMARY KEY IDENTITY(1,1),  
    full_name NVARCHAR(100) NOT NULL,  
    username NVARCHAR(50) NOT NULL UNIQUE,  
    password NVARCHAR(100) NOT NULL,  
    email NVARCHAR(100),  
    phone NVARCHAR(20),  
    address NVARCHAR(255),  
    user_type NVARCHAR(20) NOT NULL,  
    status NVARCHAR(20) DEFAULT 'active',  
    created_at DATETIME DEFAULT GETDATE()  
  
);
```

Categories Table-

```
CREATE TABLE Categories (  
  
    category_id INT PRIMARY KEY IDENTITY(1,1),  
    category_name NVARCHAR(100) NOT NULL,  
    description NVARCHAR(255)  
  
);
```

Products Table-

```
CREATE TABLE Products (  
    product_id INT PRIMARY KEY IDENTITY(1,1),  
    vendor_id INT NOT NULL,  
    category_id INT NOT NULL,  
    product_name NVARCHAR(150) NOT NULL,  
    description NVARCHAR(500),  
    price DECIMAL(10,2) NOT NULL,  
    discount_pct DECIMAL(5,2) DEFAULT 0,  
    stock_qty INT DEFAULT 0,  
    image_path NVARCHAR(255),  
    availability NVARCHAR(20) DEFAULT 'available',  
    created_at DATETIME DEFAULT GETDATE(),  
  
    FOREIGN KEY (vendor_id)  
    REFERENCES Users(user_id),
```

```
FOREIGN KEY (category_id)
REFERENCES Categories(category_id)
```

```
);
```

Cart Table

```
CREATE TABLE Cart (

    cart_id INT PRIMARY KEY IDENTITY(1,1),
    customer_id INT NOT NULL,
    product_id INT NOT NULL,
    quantity INT NOT NULL DEFAULT 1,
    added_at DATETIME DEFAULT GETDATE(),

    FOREIGN KEY (customer_id)
    REFERENCES Users(user_id),

    FOREIGN KEY (product_id)
    REFERENCES Products(product_id)

);
```

Order Table-

```
CREATE TABLE Orders (

    order_id INT PRIMARY KEY IDENTITY(1,1),
    customer_id INT NOT NULL,
    product_id INT NOT NULL,
    vendor_id INT NOT NULL,
    quantity INT NOT NULL,
    total_price DECIMAL(10,2) NOT NULL,
    order_status NVARCHAR(30) DEFAULT 'Placed',
    payment_method NVARCHAR(30),
    delivery_address NVARCHAR(255),
    order_date DATETIME DEFAULT GETDATE(),
    updated_at DATETIME DEFAULT GETDATE(),

    FOREIGN KEY (customer_id)
    REFERENCES Users(user_id),

    FOREIGN KEY (product_id)
    REFERENCES Products(product_id),

    FOREIGN KEY (vendor_id)
    REFERENCES Users(user_id)

);
```

Payment Table-

```
CREATE TABLE Payments (

    payment_id INT PRIMARY KEY IDENTITY(1,1),
    order_id INT NOT NULL,
    customer_id INT NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_method NVARCHAR(30) NOT NULL,
    transaction_ref NVARCHAR(100),
    payment_status NVARCHAR(20) DEFAULT 'completed',
```

```

        payment_date DATETIME DEFAULT GETDATE(),

        FOREIGN KEY (order_id)
        REFERENCES Orders(order_id),

        FOREIGN KEY (customer_id)
        REFERENCES Users(user_id)

    );

```

Reviews Table-

```

CREATE TABLE Reviews (

    review_id INT PRIMARY KEY IDENTITY(1,1),
    customer_id INT NOT NULL,
    product_id INT NOT NULL,
    order_id INT NOT NULL,
    rating INT NOT NULL,
    comment NVARCHAR(500),
    review_date DATETIME DEFAULT GETDATE(),

    FOREIGN KEY (customer_id)
    REFERENCES Users(user_id),

    FOREIGN KEY (product_id)
    REFERENCES Products(product_id),

    FOREIGN KEY (order_id)
    REFERENCES Orders(order_id)

);

```

Users Insert Data-

```

INSERT INTO Users
(full_name, username, password, email, phone,
address, user_type, status)

```

```

VALUES
('Muhammad Taki Ahmed','taki','taki1234',
'taki@needbridge.com','01700000001',
'Dhaka','superadmin','active');

```

```

INSERT INTO Users
VALUES
('Dr. Md. Iftekharul Mobin',
'iftekharul','sirmobin',
'mobin@needbridge.com',
'01700000002','Dhaka',
'admin','active',GETDATE());

```

```

INSERT INTO Users
VALUES
('Hasibul Hasan',
'hasibul','hasib1234',
'hasibul@needbridge.com',
'01700000003','Dhaka',
'admin','active',GETDATE());

```

```
INSERT INTO Users
VALUES
('Md. Touhidul Islam',
'touhidul','touhi1234',
'touhidul@needbridge.com',
'01700000004','Dhaka',
'admin','active',GETDATE());
```

```
INSERT INTO Users
VALUES
('Mohtasim Mohammad',
'mohtasim','mohta1234',
'mohtasim@needbridge.com',
'01700000005','Dhaka',
'admin','active',GETDATE());
```

Categories Insert Data-

```
INSERT INTO Categories
(category_name, description)
```

```
VALUES
('Electronics',
'Phones, laptops, gadgets'),
('Clothing',
'Shirts, pants, shoes'),
('Food & Grocery',
'Daily food items'),
('Home & Living',
'Furniture, decor'),
('Beauty & Health',
'Skincare, medicine'),
('Sports',
'Equipment, clothing'),
('Books & Stationery',
'Books, pens, notebooks'),
('Automotive',
'Car parts, accessories');
```

Vendor & Customer Data-

```
INSERT INTO Users
(full_name, username, password,
email, phone, address,
user_type, status)
```

```
VALUES
```

```
('Test Vendor',
'testvendor',
'vendor123',
'vendor@test.com',
'01700000010',
'Dhaka',
'vendor',
'active'),
```

```
('Test Customer',
'testcustomer',
'test123',
'customer@test.com',
```

```
'01700000020',  
'Dhaka',  
'customer',  
'active');
```

Product Insert Data-

```
INSERT INTO Products
```

```
(vendor_id,  
category_id,  
product_name,  
description,  
price,  
discount_pct,  
stock_qty,  
availability)
```

```
VALUES
```

```
(  
6,  
1,  
'Samsung Galaxy A55',  
'8GB RAM, 256GB Storage, AMOLED Display',  
45000,  
10,  
20,  
'available'  
);
```

Display Queries:-

```
SELECT * FROM Users;
```

```
SELECT * FROM Categories;
```

```
SELECT * FROM Products;
```

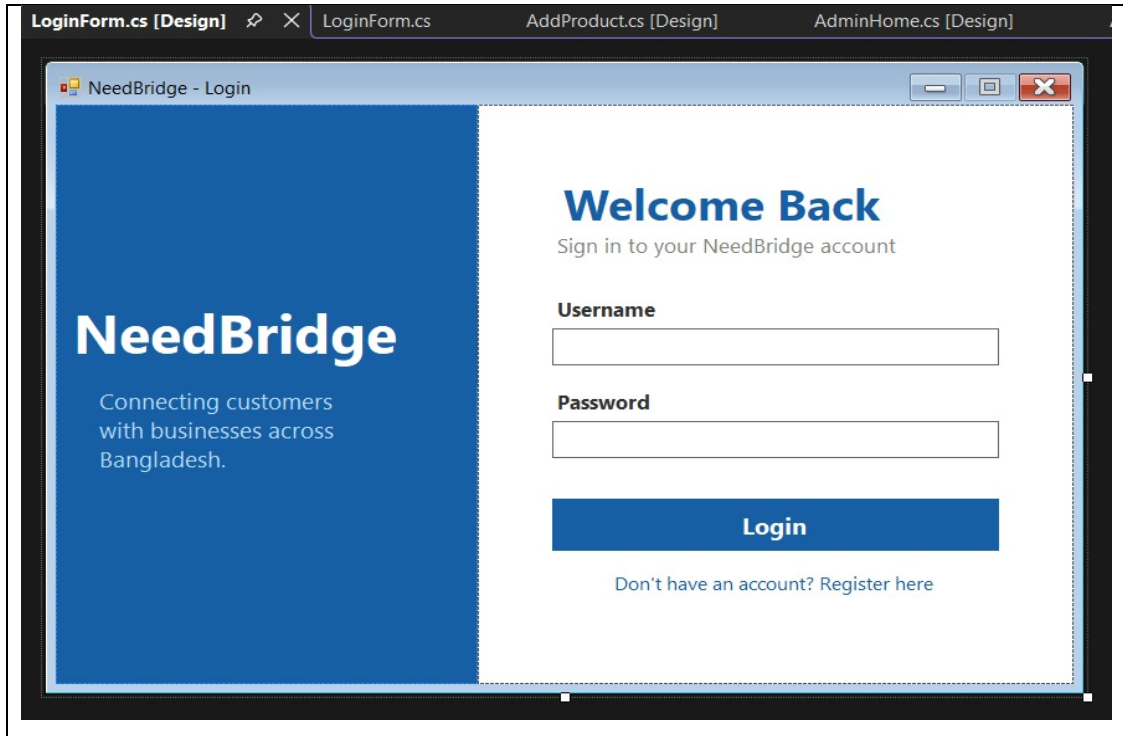
```
SELECT * FROM Cart;
```

```
SELECT * FROM Orders;
```

```
SELECT * FROM Payments;
```

```
SELECT * FROM Reviews;
```

Project Screenshot- Log in Panel



The screenshot shows a web browser window with three tabs: 'LoginForm.cs [Design]', 'AddProduct.cs [Design]', and 'AdminHome.cs [Design]'. The active tab is 'LoginForm.cs [Design]'. The browser window title is 'NeedBridge - Login'. The login panel has a blue sidebar on the left with the 'NeedBridge' logo and the tagline 'Connecting customers with businesses across Bangladesh.' The main content area is white and features a 'Welcome Back' message, a prompt to 'Sign in to your NeedBridge account', and a login form with fields for 'Username' and 'Password', a 'Login' button, and a link to 'Register here' for users without an account.

NeedBridge
Connecting customers with businesses across Bangladesh.

Welcome Back
Sign in to your NeedBridge account

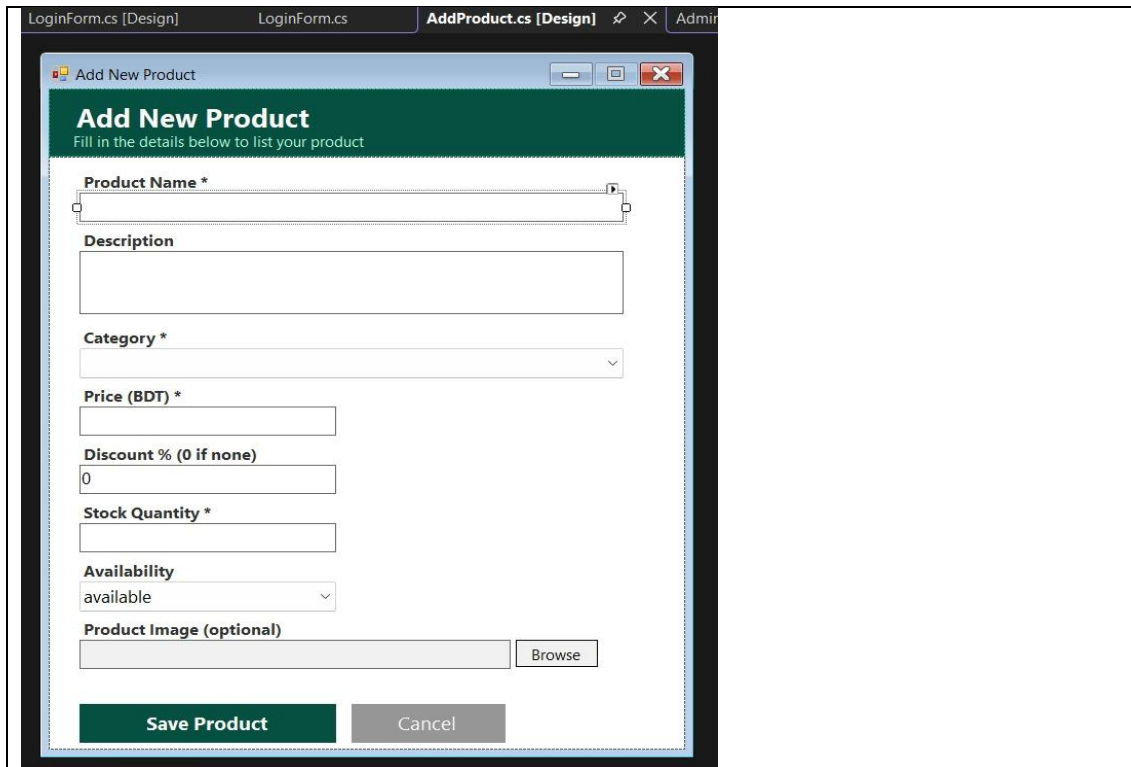
Username

Password

Login

[Don't have an account? Register here](#)

Product Adding Dashboard-



The screenshot shows a web browser window with three tabs: 'LoginForm.cs [Design]', 'LoginForm.cs', and 'AddProduct.cs [Design]'. The active tab is 'AddProduct.cs [Design]'. The browser window title is 'Add New Product'. The dashboard has a green header with the title 'Add New Product' and the instruction 'Fill in the details below to list your product'. The form contains several fields: 'Product Name *' (text input), 'Description' (text area), 'Category *' (dropdown menu), 'Price (BDT) *' (text input), 'Discount % (0 if none)' (text input with value '0'), 'Stock Quantity *' (text input), 'Availability' (dropdown menu with value 'available'), and 'Product Image (optional)' (text input with a 'Browse' button). At the bottom, there are 'Save Product' and 'Cancel' buttons.

Add New Product
Fill in the details below to list your product

Product Name *

Description

Category *

Price (BDT) *

Discount % (0 if none)

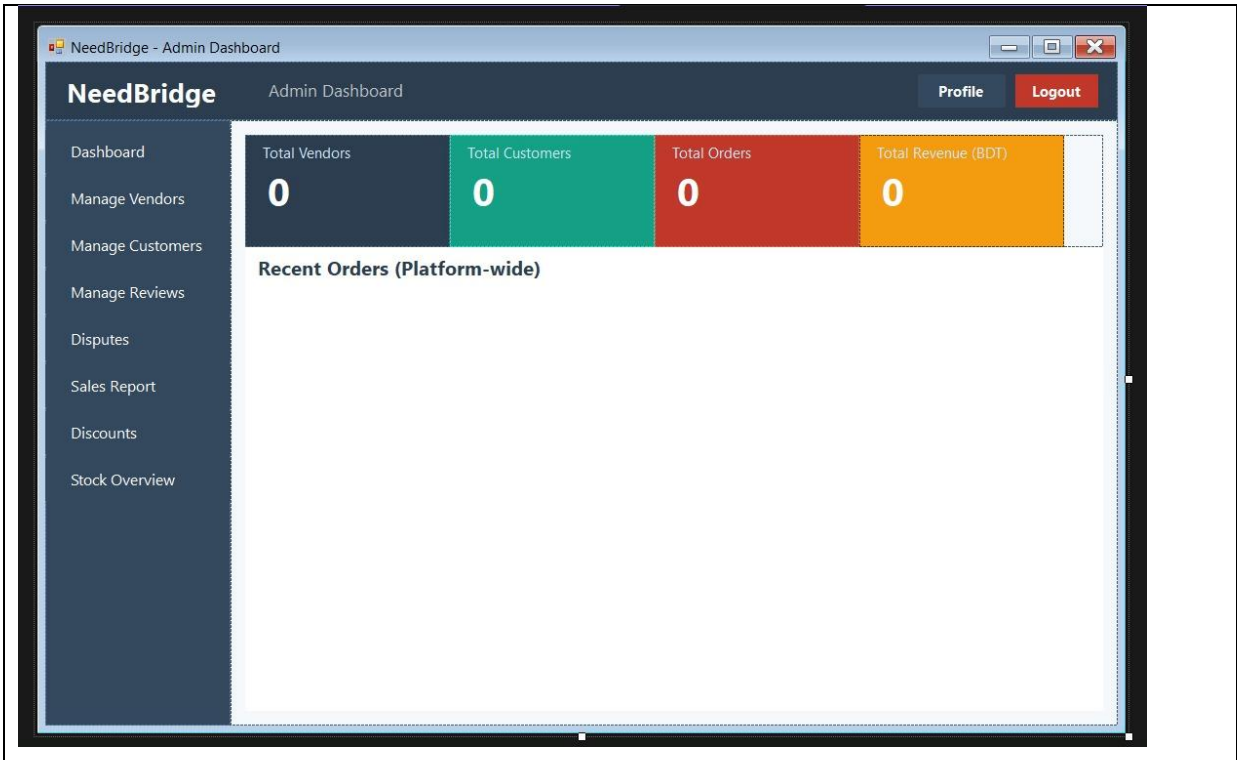
Stock Quantity *

Availability

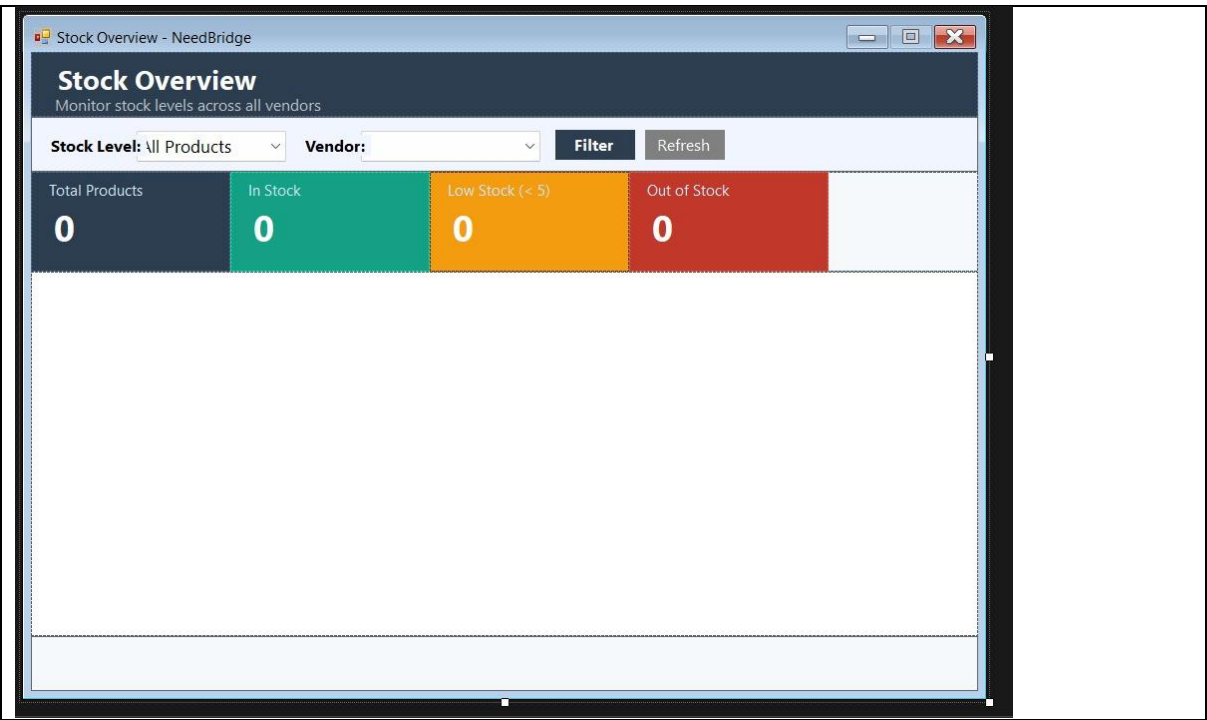
Product Image (optional)

Save Product

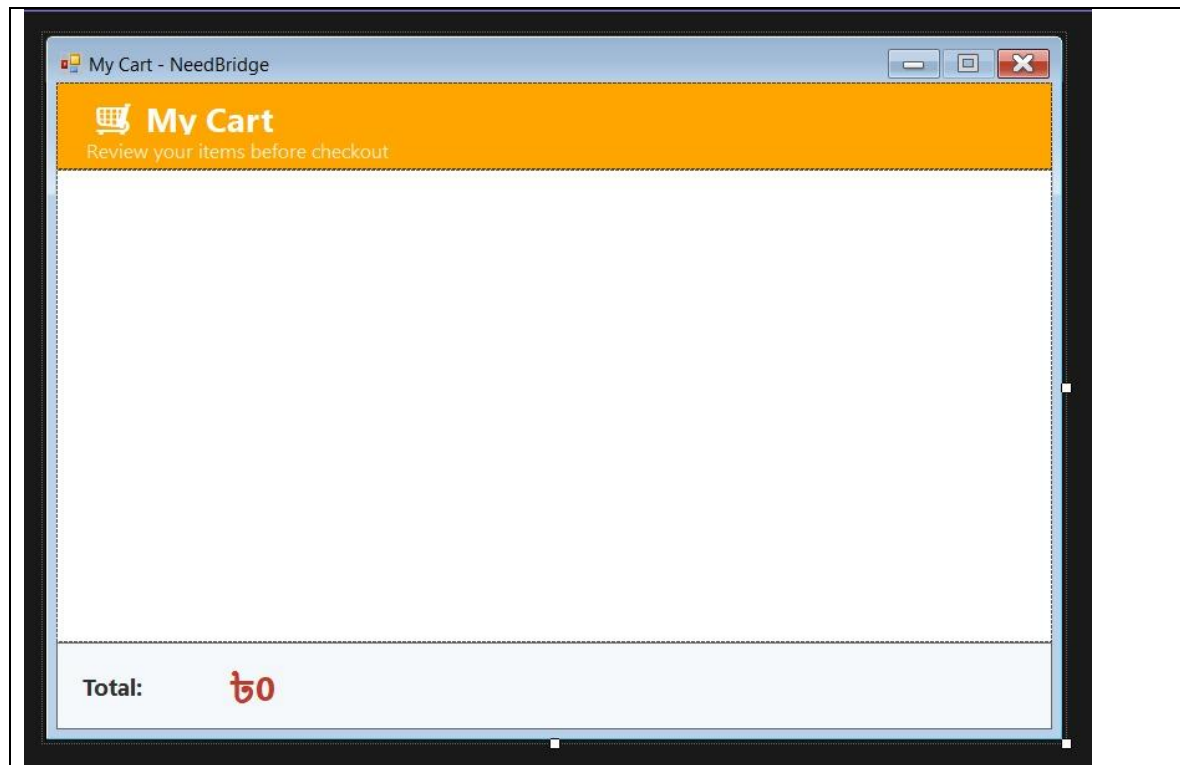
Admin Dashboard-



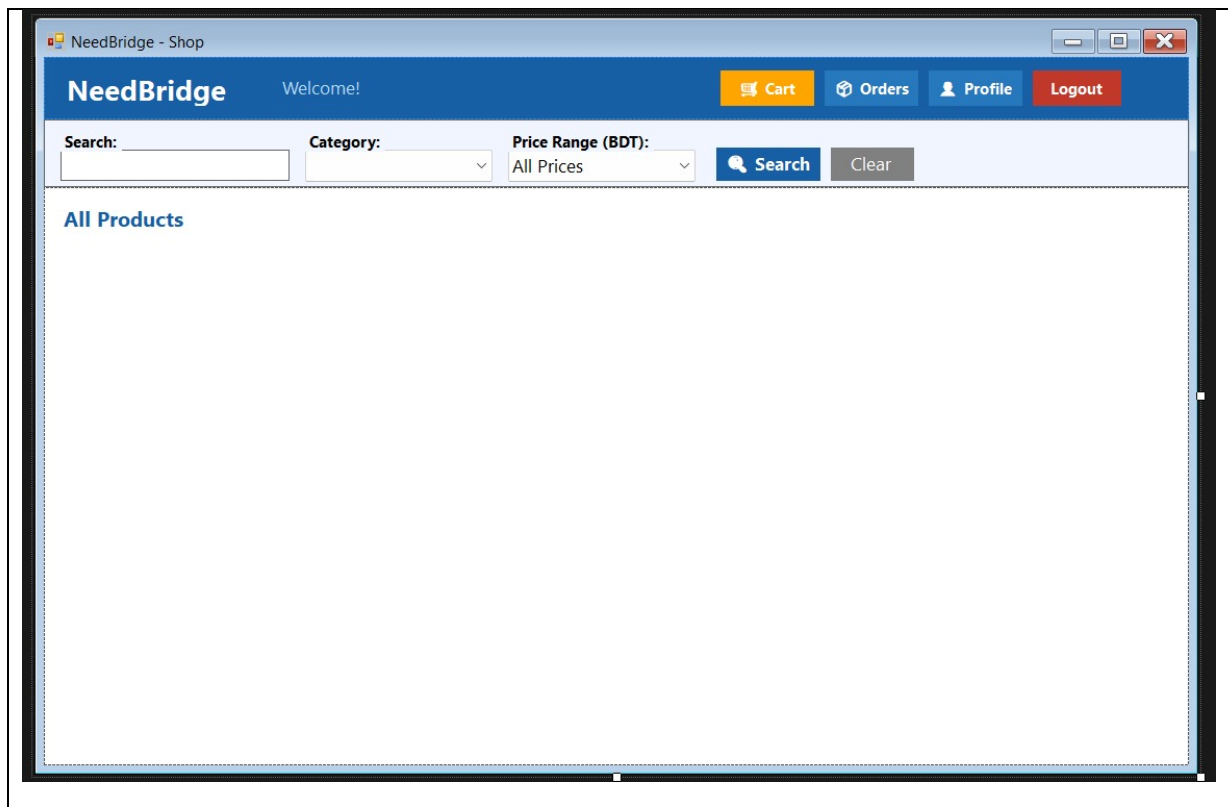
Stock Overview-



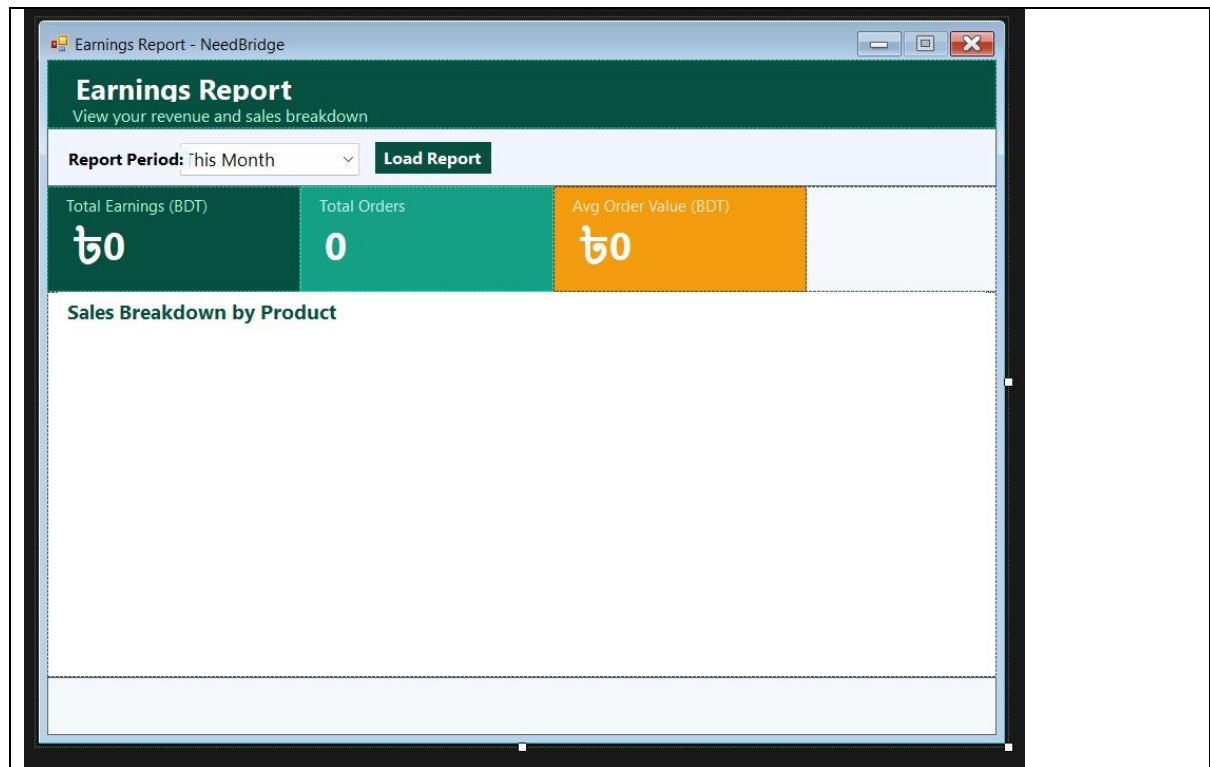
My Cart-



Shop:-



Earning Report-



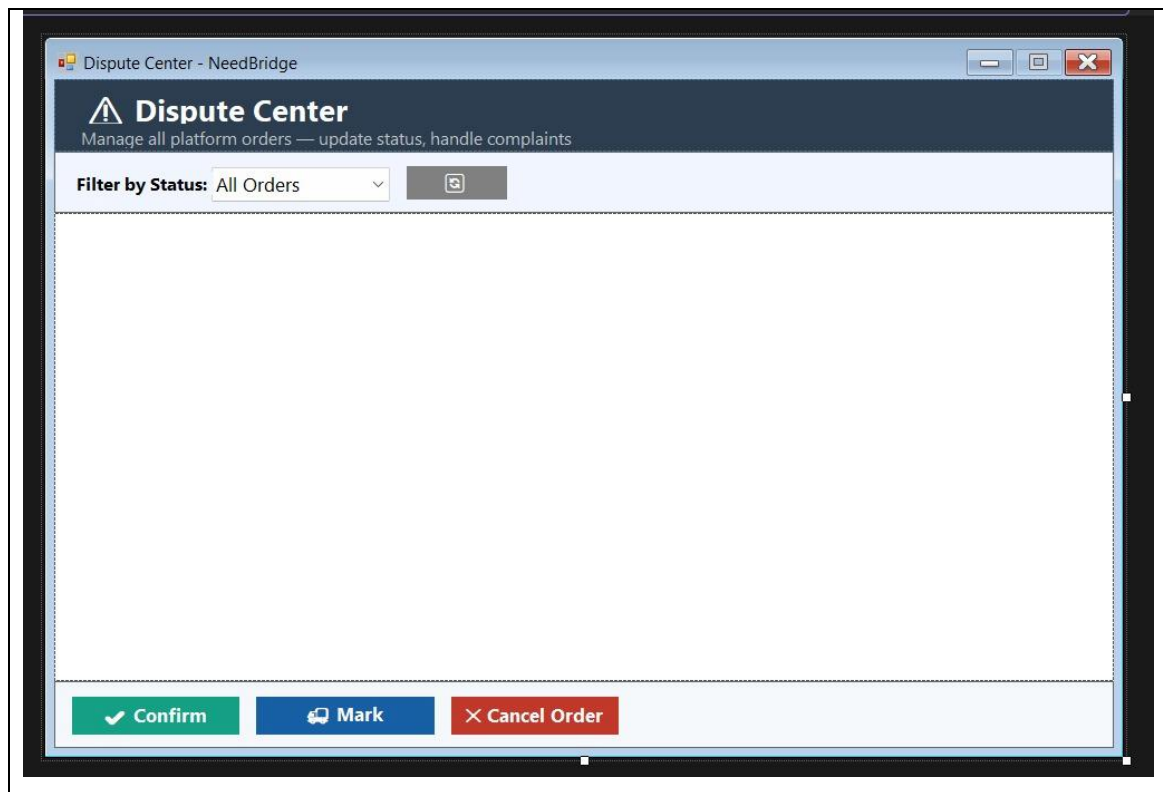
Manage Discount-

The screenshot shows a web application window titled "Manage Discounts - NeedBridge". The main heading is "Manage Discounts" with the subtitle "Set or remove discounts on any product". Below this, there is a search bar labeled "Search:" and a "Category:" dropdown menu. To the right of the search bar are "Search" and "Refresh" buttons. The main area of the interface is empty. At the bottom, there is a "Discount %:" input field with the value "10", and two buttons: "Apply Discount" and "Remove Discount".

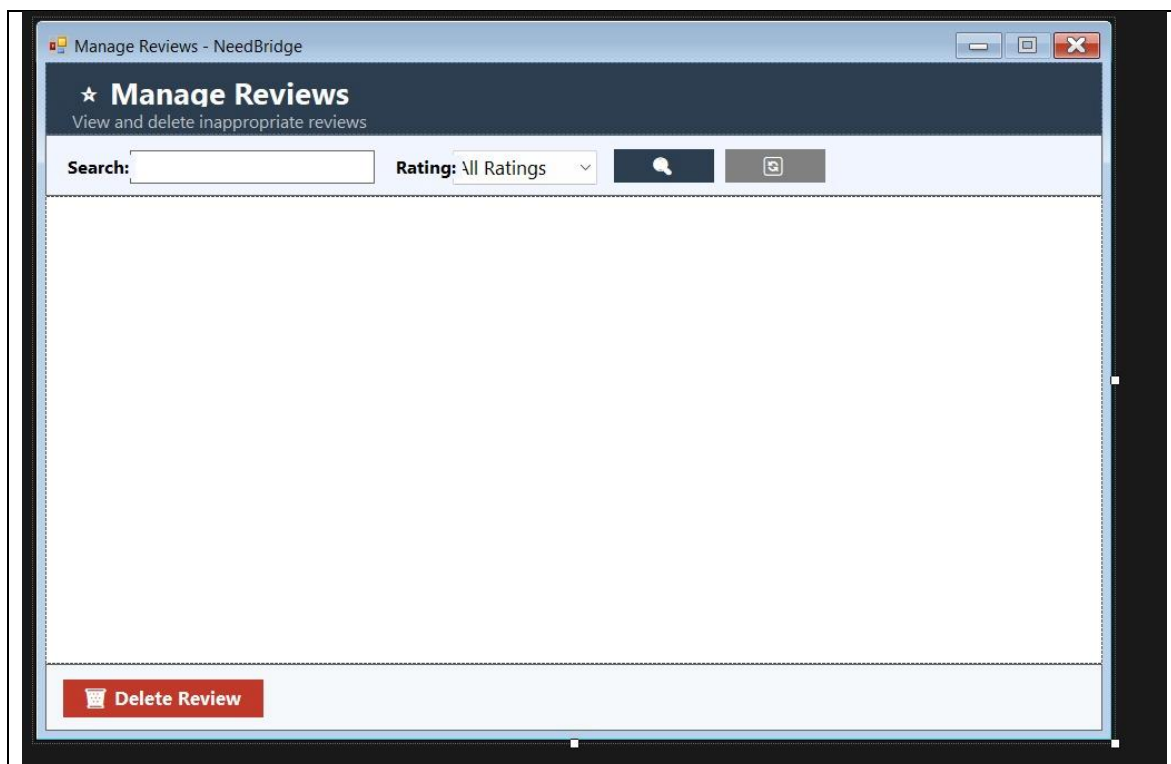
Search: Category:

Discount %:

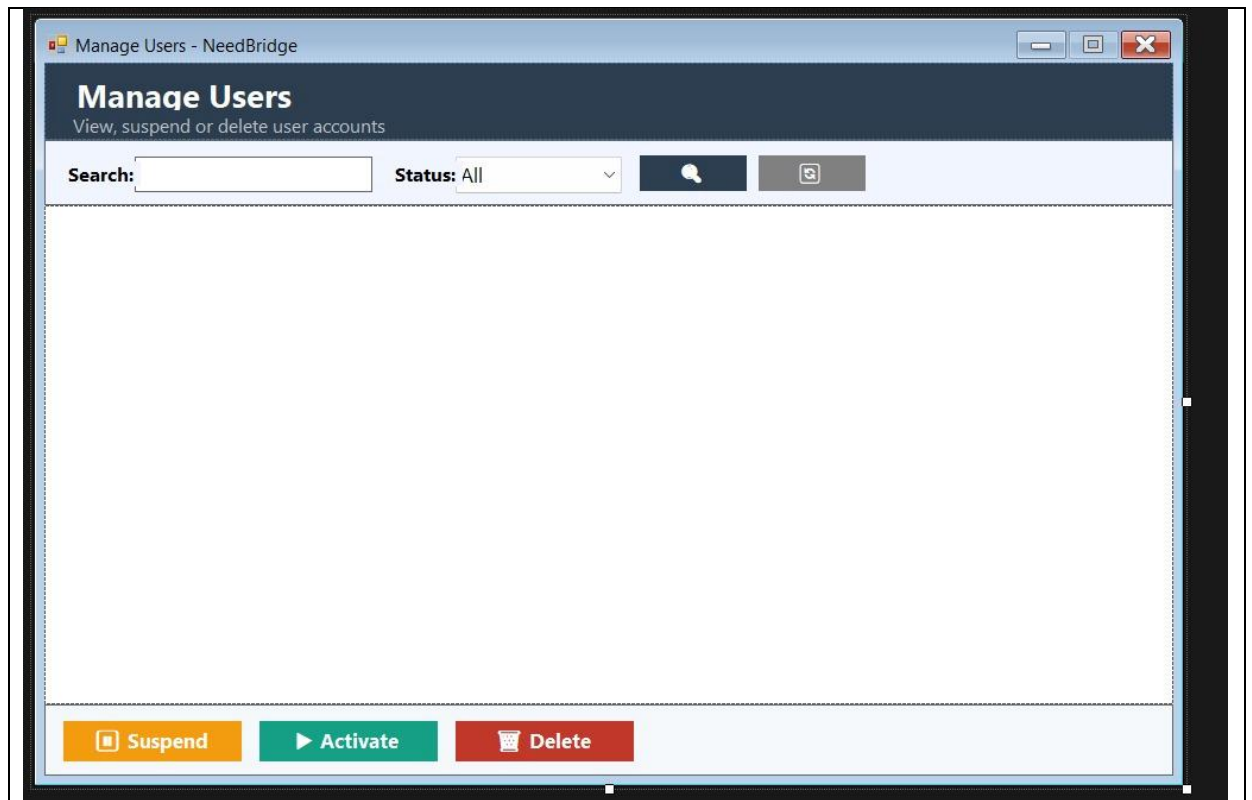
Display Center-



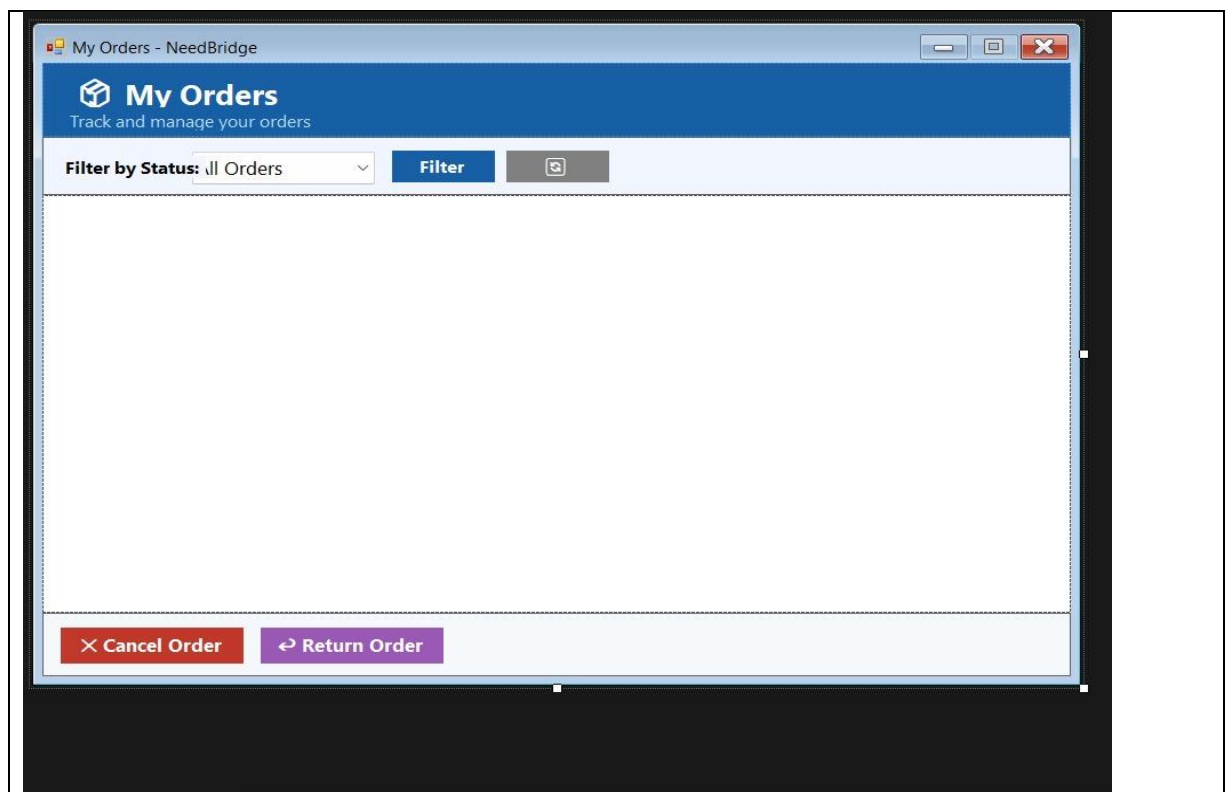
Manage Review:-



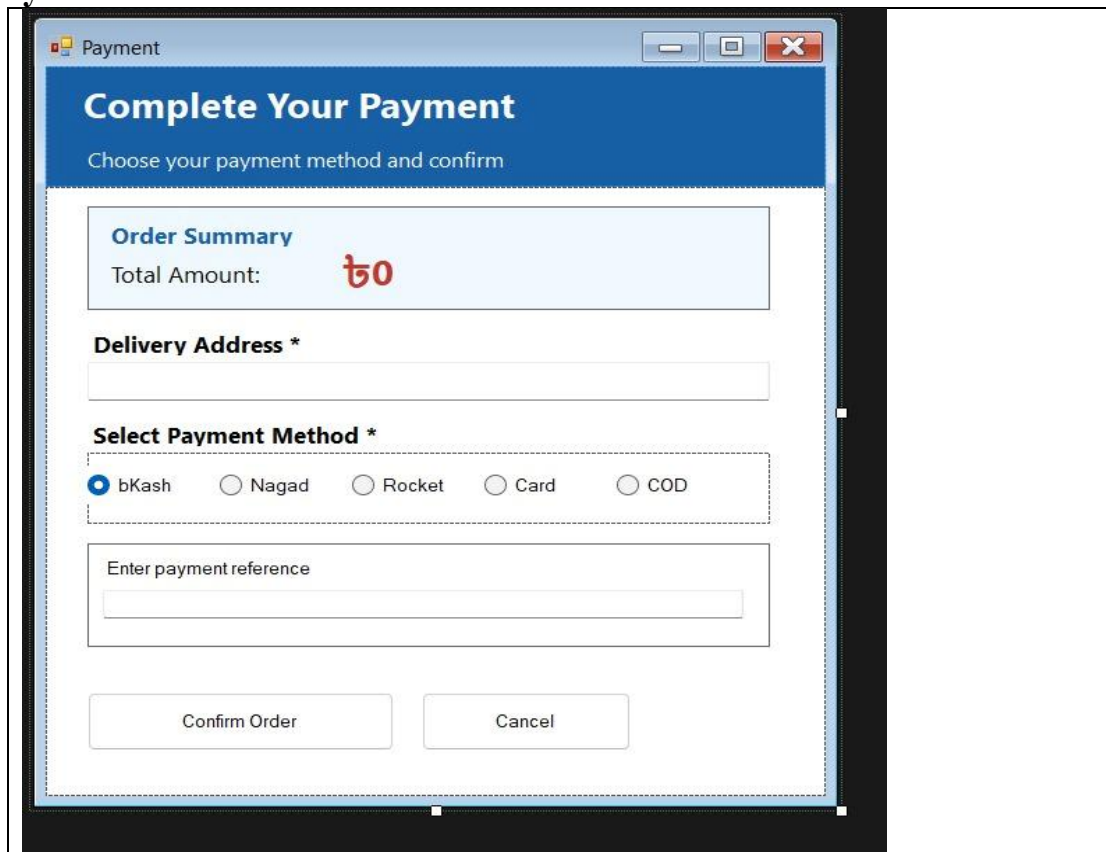
Manage Users:



My Orders:-

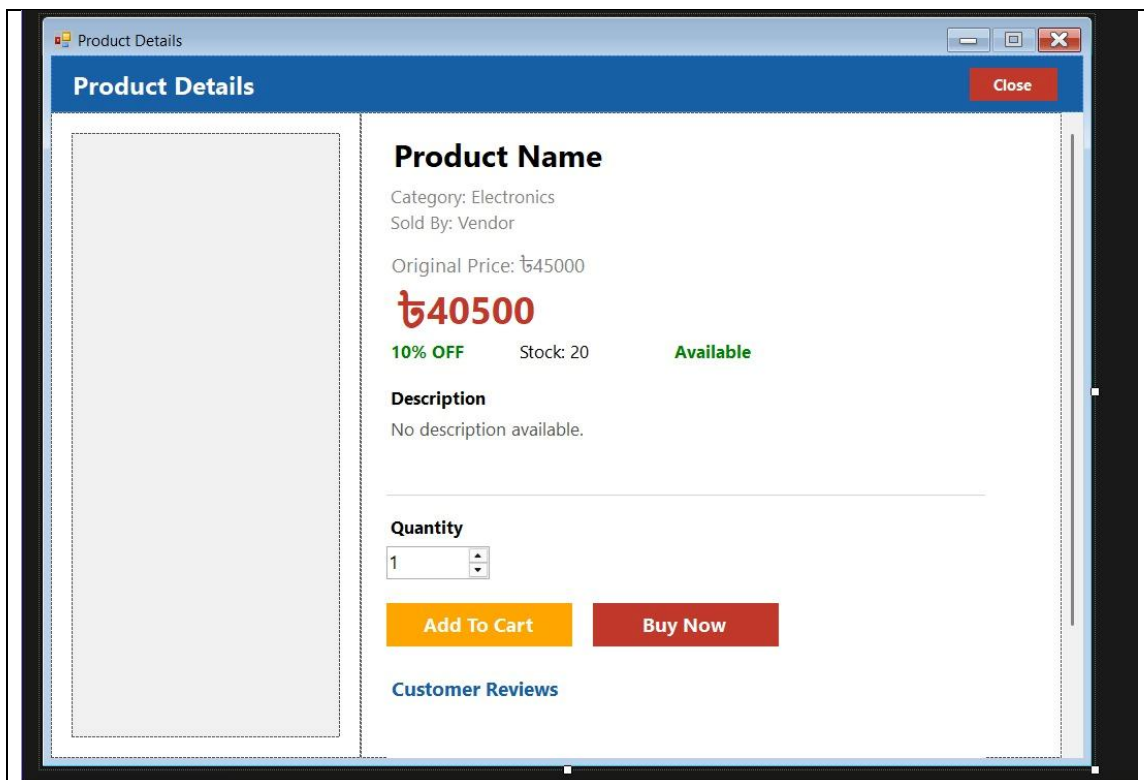


Payment Method:-



A screenshot of a web application window titled "Payment". The window has a blue header bar with the title "Complete Your Payment" and a subtitle "Choose your payment method and confirm". Below the header, there is a light blue box labeled "Order Summary" containing the text "Total Amount: ₹0". Underneath, there is a section for "Delivery Address *" with an empty text input field. This is followed by a "Select Payment Method *" section with five radio button options: "bKash" (selected), "Nagad", "Rocket", "Card", and "COD". Below the payment methods is a section labeled "Enter payment reference" with an empty text input field. At the bottom of the window, there are two buttons: "Confirm Order" and "Cancel".

Product Sample:-



A screenshot of a web application window titled "Product Details". The window has a blue header bar with the title "Product Details" and a "Close" button. The main content area is divided into two columns. The left column contains a large, empty rectangular box. The right column displays product information: "Product Name", "Category: Electronics", "Sold By: Vendor", "Original Price: ₹45000", and a large red price tag "₹40500". Below the price, it shows "10% OFF", "Stock: 20", and "Available". There is a "Description" section with the text "No description available." and a "Quantity" section with a dropdown menu set to "1". At the bottom of the right column, there are two buttons: "Add To Cart" (orange) and "Buy Now" (red). Below these buttons is a link for "Customer Reviews".

Customer Profile:-

My Profile
View and update your account details

Full Name

Username (cannot change)

Email

Phone Number

Address

Account Type

Member Since

Change Password (optional)

Current Password

New Password

Confirm New Password

List

Register:-

NeedBridge
Join thousands of vendors and customers across Bangladesh.

Create Account
Fill in the details below to get started

Full Name

Username

Email

Phone Number

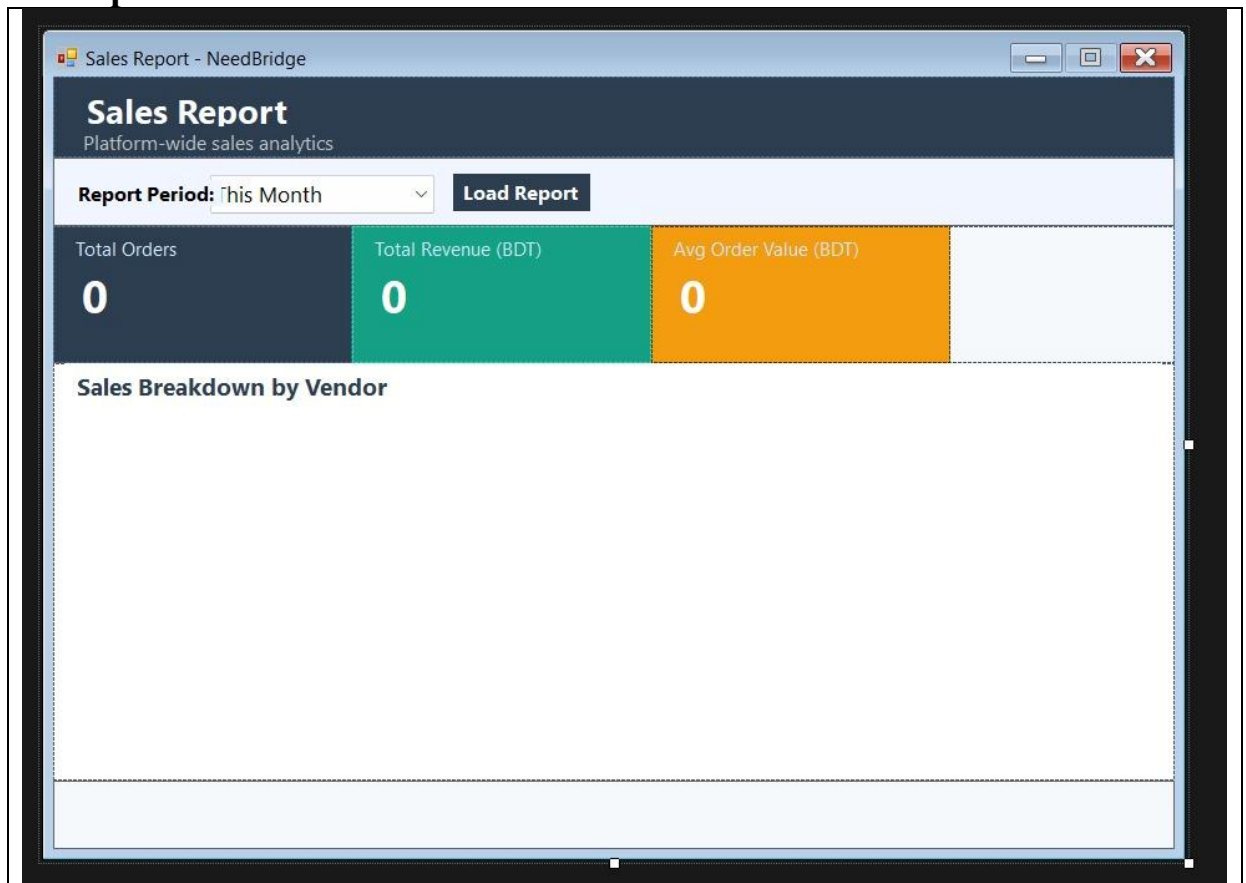
Address

Password

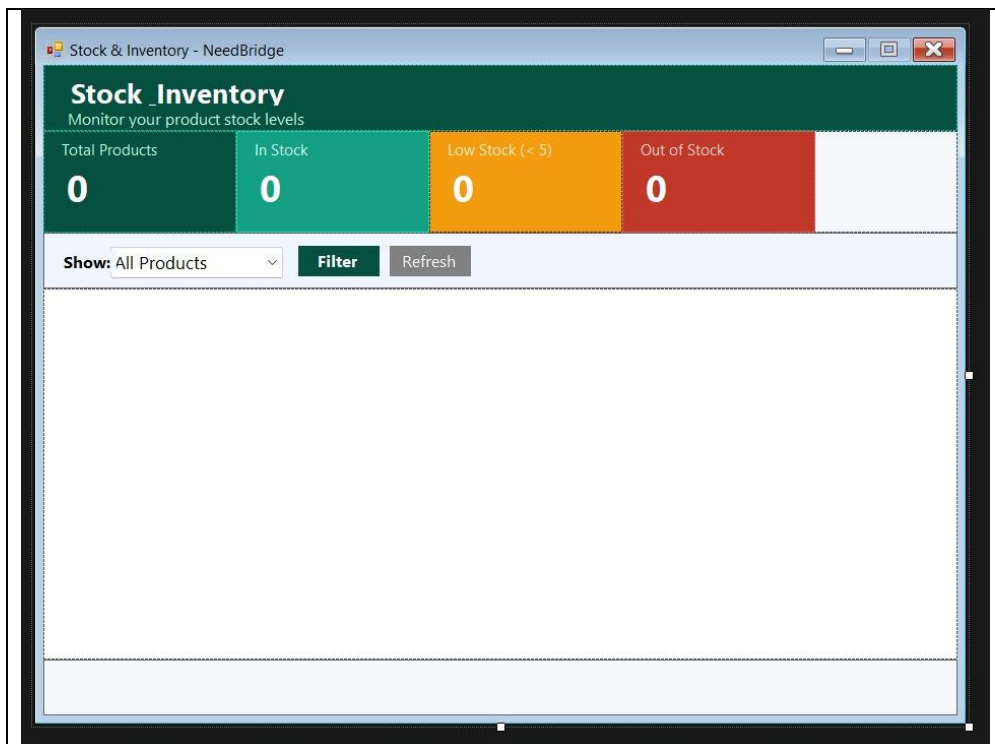
Register As

Create Account

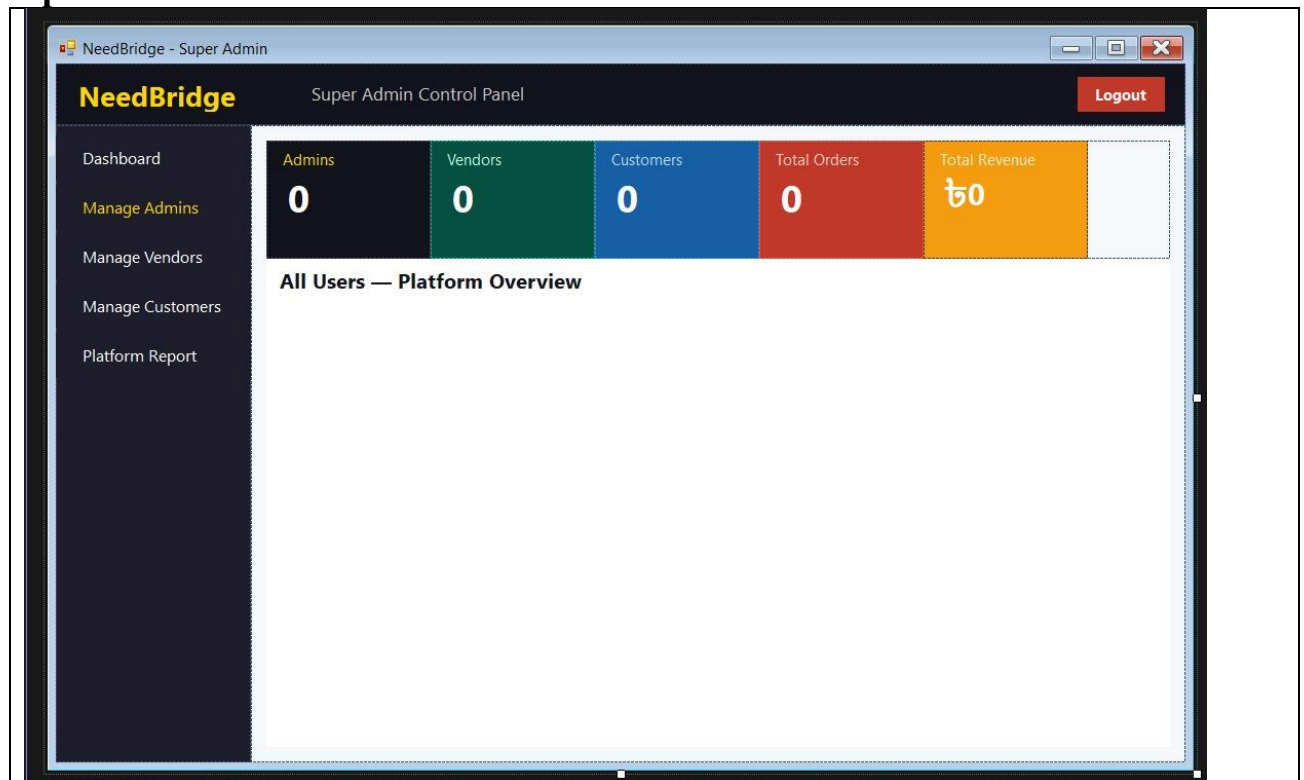
Sales Report:-



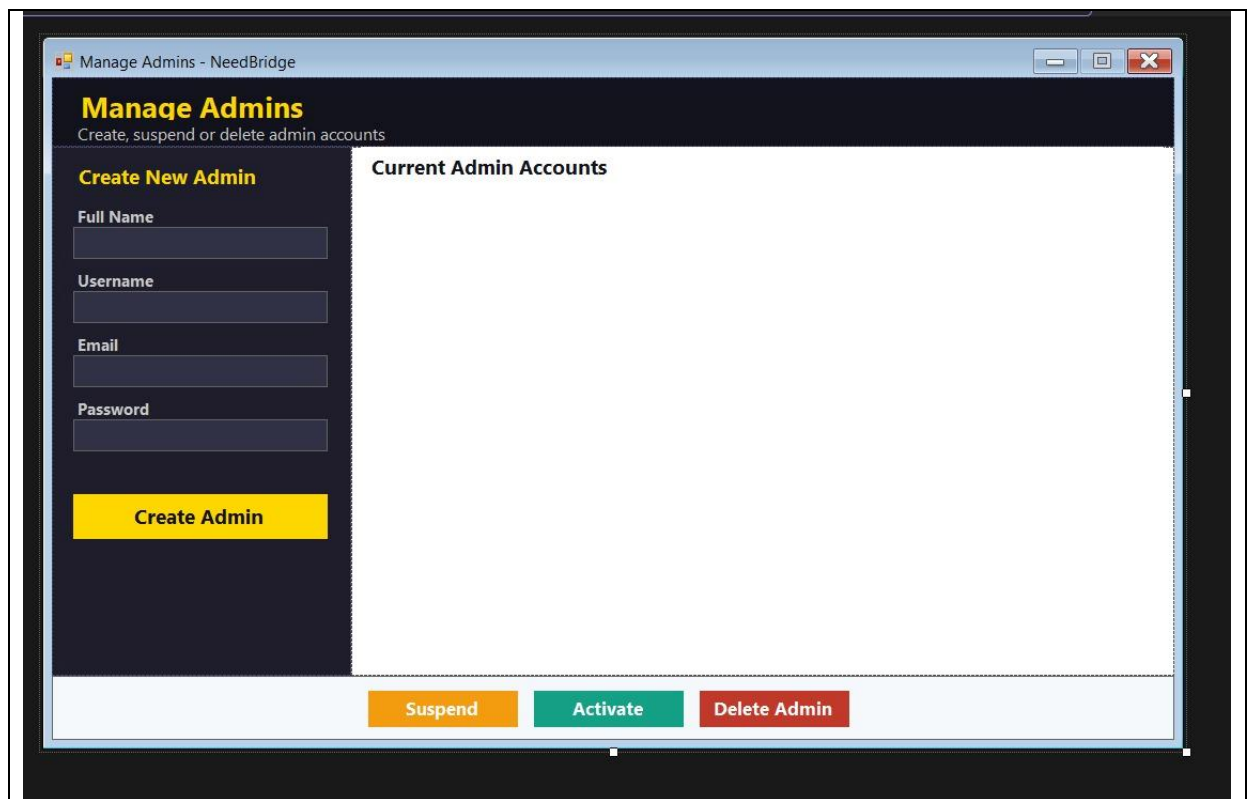
Stock & Inventory:-



Super Admin Control Panel:-



Manage Admins:-



My Orders:-

The screenshot shows a web browser window titled "My Orders - NeedBridge". The main heading is "My Orders" with a subtitle "View and update delivery status of your orders". Below this, there is a "Filter by Status:" dropdown menu set to "All Orders", a green "Filter" button, and a grey "Refresh" button. The central area is a large empty box for displaying orders. At the bottom, there is an "Update Status:" dropdown menu set to "Confirmed" and a green "Update Status" button.

My Product:-

The screenshot shows a web browser window titled "My Products - NeedBridge". The main heading is "My Products" with a subtitle "Manage your product listings". Below this, there is a "Search:" input field, a green "Search" button, a grey "Refresh" button, and a blue "+ Add" button. The central area is a large empty box for displaying products. On the right side, there is a sidebar titled "Edit Selected Product" containing several input fields: "Product Name", "Price (BDT)", "Discount %", and "Stock Quantity". Below these is an "Availability" dropdown menu set to "available". At the bottom of the sidebar, there are two buttons: a green "Save Changes" button and a red "Delete Product" button.

Review :-

The screenshot shows a web browser window titled "Write Review - NeedBridge". The main heading is "Write a Review" with the subtitle "Share your experience with this product". Below this, it says "Loading product...". The "Your Rating *" section features five radio buttons for 1 Star, 2 Stars, 3 Stars, 4 Stars, and 5 Stars. The 5 Stars option is selected, and the word "Excellent!" is displayed in green. Below the rating is a text area for "Your Comment". At the bottom, there are two buttons: "Submit Review" (blue) and "Cancel" (grey).

Write Review - NeedBridge

Write a Review

Share your experience with this product

Loading product...

Your Rating *

☐ 1 Star ☐ 2 Stars ☐ 3 Stars ☐ 4 Stars ☒ 5 Stars

Excellent!

Your Comment

Submit Review **Cancel**

Vendor Dashboard:-

The screenshot shows a web browser window titled "NeedBridge - Vendor Dashboard". The page has a dark green sidebar with the "NeedBridge" logo and a list of menu items: Dashboard, My Products, Add Product, My Orders, Stock, and Earnings. The main content area has a header with "Vendor Dashboard" and links for "Profile" and "Logout". Below the header, there are four colored boxes showing statistics: "Total Products" (0, blue), "Total Orders" (0, green), "Total Earnings (BDT)" (0, red), and "Low Stock Items" (0, purple). Below these boxes is a section titled "Recent Orders" with a large empty white area.

NeedBridge - Vendor Dashboard

NeedBridge Vendor Dashboard Profile Logout

Dashboard My Products Add Product My Orders Stock Earnings

Total Products	Total Orders	Total Earnings (BDT)	Low Stock Items
0	0	0	0

Recent Orders